

TENDRA-H1

Tendon-driven dexterous robotic hand with an exposed CF-nylon skeleton, opposable thumb and 2-DOF wrist. Every figure on this page is computed from the parametric model itself.

25

JOINTS

10

ACTUATORS

457 g

EST. MASS

\$1,441

BOM COST

309

PARTS



GENERAL

Envelope (L × W × D)	298 × 152 × 69 mm
Wrist centre to middle tip	190 mm
Fingers	4 + opposable thumb
Unique line items	64
Structure	PA12-CF links (MJF), Ti-6Al-4V yokes & gimbal, 7075-T6 frame
Joint hardware	Ø3 mm 440C pins, flanged ball bearings, M2 12.9 screws
Sensing	Capacitive tactile pad per fingertip, magnetic encoder per actuator

ACTUATION

Method	Tendon-driven (Dyneema Ø1.0 over grooved pulleys), under-actuated coupled flexion, elastic extension
Actuators	10 × Ø9.5 mm coreless DC gearmotor, 136:1
Tendon	Dyneema Ø1.0 braid over grooved 440C pulleys
Grasp presets	Open · Fist · Pinch · Point · Power Grip · OK Sign

MASS BY ASSEMBLY

Forearm	227 g
Palm	46 g
Middle	42 g
Index	38 g
Ring	35 g
Thumb	29 g
Little	24 g
Wrist	15 g

RANGE OF MOTION

JOINT	MIN	MAX	QTY	DRIVEN BY
MCP abduction	-20°	+20°	×4	Finger Spread
MCP flexion	-10°	+90°	×4	Finger Flexor
PIP flexion	-5°	+110°	×4	Finger Flexor (coupled)
DIP flexion	-10°	+80°	×4	Finger Flexor (coupled)
CMC (cupping)	+0°	+15°	×2	Palm Cup
Thumb CMC flexion	-10°	+65°	×1	Thumb Opposition
Thumb CMC axial rotation	-20°	+55°	×1	Thumb Opposition (coupled)
Thumb CMC abduction	-30°	+75°	×1	Thumb Opposition (coupled)
Thumb MCP flexion	-10°	+60°	×1	Thumb Flexor
Thumb IP flexion	-15°	+80°	×1	Thumb Flexor (coupled)
Wrist pitch	-60°	+70°	×1	Wrist Pitch
Wrist yaw	-20°	+30°	×1	Wrist Yaw

TOP COST DRIVERS

ITEM	QTY	G	USD
Coreless DC gearmotor Ø9.5, 136:1, magnetic encoder	10	135.0	\$420
10-channel motor driver + sensor hub PCB	1	4.8	\$120
Capacitive tactile sensor pad	5	0.4	\$70
Wrist universal-joint cross, Ti-6Al-4V	1	8.4	\$58
Grooved tendon pulley, 440C	14	6.5	\$49
Fingertip cap, Ti-6Al-4V	5	3.7	\$45

ORTHOGRAPHIC VIEWS

VERIFIED · 0 INTERFERENCES · NO FLOATING PARTS

